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CS23532-COMPUTER NETWORKS-LAB MANUAL
PRACTICAL – 15

AIM:

To analyse the log files of various types of servers (web, proxy and FTP servers) using the Webalizer web server log analysis tool and to study the usage statistics generated by it.

ALGORITHM / PROCEDURE:

1. Install Webalizer in Linux using `sudo apt install webalizer apache2`.
2. Locate the log files of the servers to be analysed (Apache, Nginx, IIS, FTP, Squid) as given in the table below.
3. Edit the configuration file `/etc/webalizer/webalizer.conf` and set `LogFile`, `LogType`, `OutputDir` and `HostName`.
4. Run Webalizer for each server. The option `-n` gives the host name for the report, `-o` gives the output folder and `-F` gives the log type (clf, w3c, ftp, squid).
5. Open the generated `usage_YYYYMM.html` file in a web browser (`http://localhost/webalizer/apache/`) to see the report.
6. Study the monthly summary, the daily graph, the top URLs, the top clients, the user agents and the HTTP response codes.
7. Compare the results of the different servers and note down the inference.

Servers and their log formats

Server type	Log file	Log format	Webalizer option
Apache web server	`/var/log/apache2/access.log`	Common Log Format (CLF)	`-F clf`
Nginx web server	`/var/log/nginx/access.log`	CLF compatible (combined)	`-F clf`
IIS web server	`C:\inetpub\logs\LogFiles\W3SVC1\u_ex261008.log`	W3C extended	`-F w3c`
FTP server (vsftpd)	`/var/log/xferlog`	wu-ftpd xferlog	`-F ftp`
Squid proxy server	`/var/log/squid/access.log`	Squid native	`-F squid`

webalizer.conf (important lines)

```
LogFile         /var/log/apache2/access.log
LogType         clf
OutputDir       /var/www/html/webalizer/apache
HostName        apache-server
PageType        htm*
PageType        php
PageType        cgi
HideSite        localhost
```

Sample log records

```
Apache (CLF):
192.168.1.25 - - [08/Oct/2026:10:41:07 +0530] "GET /index.html HTTP/1.1" 200 4521
192.168.1.12 - - [08/Oct/2026:10:41:09 +0530] "GET /css/style.css HTTP/1.1" 304 -
10.0.2.2 - - [08/Oct/2026:10:42:15 +0530] "GET /missing.html HTTP/1.1" 404 492
IIS (W3C):
#Fields: date time s-ip cs-method cs-uri-stem sc-status sc-bytes
2026-10-08 05:12:44 192.168.1.30 GET /default.aspx 200 6120
```

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DIAGRAM:

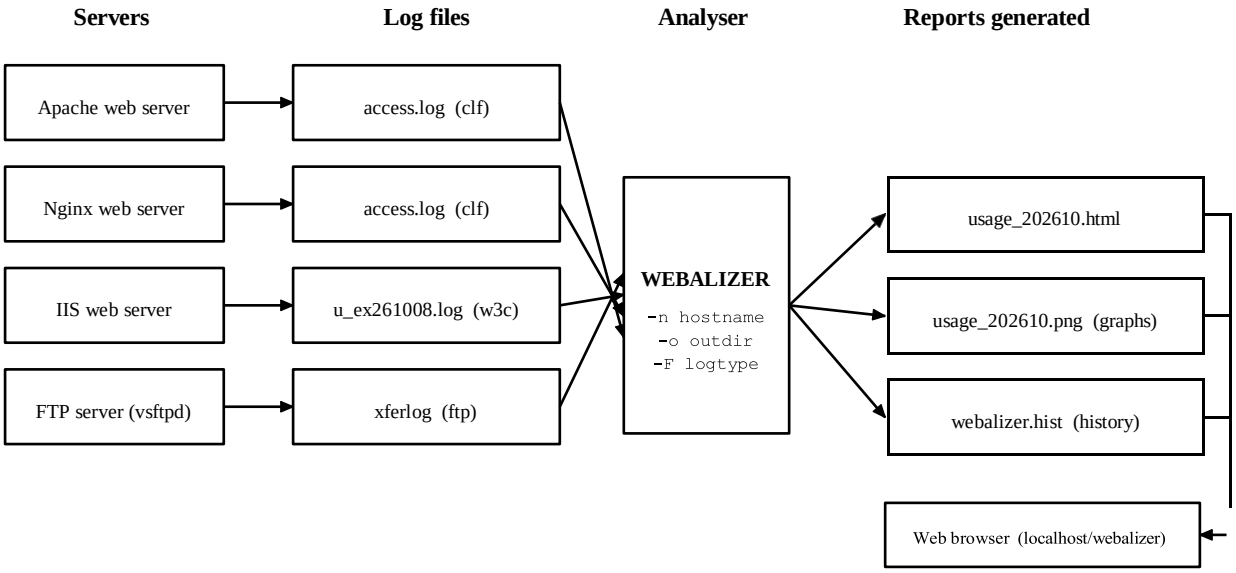


Fig 1: Log analysis of different servers using Webalizer

OUTPUT:

```
user@user-VirtualBox:~$ sudo webalizer -n apache-server -o /var/www/html/webalizer/apache -F clf /va

  r/log/apache2/access.log
Webalizer V2.23-08 (Linux 5.15.0-91-generic) English
Using logfile /var/log/apache2/access.log (clf)
Creating output in /var/www/html/webalizer/apache
Hostname for reports is 'apache-server'
History file not found...
Generating report for October 2026
Generating summary report
Saving history information...
1243 records in 0 seconds
user@user-VirtualBox:~$ sudo webalizer -n nginx-server -o /var/www/html/webalizer/nginx -F clf /var/

  log/nginx/access.log
Using logfile /var/log/nginx/access.log (clf)
Creating output in /var/www/html/webalizer/nginx
Generating report for October 2026
Generating summary report
Saving history information...
856 records in 0 seconds
user@user-VirtualBox:~$ sudo webalizer -n iis-server -o /var/www/html/webalizer/iis -F w3c u_ex26100

  8.log
Using logfile u_ex261008.log (w3c)
Creating output in /var/www/html/webalizer/iis
Generating report for October 2026
Generating summary report
Saving history information...
522 records in 0 seconds
user@user-VirtualBox:~$ sudo webalizer -n ftp-server -o /var/www/html/webalizer/ftp -F ftp /var/log/

  xferlog
Using logfile /var/log/xferlog (ftp)
Generating report for October 2026
64 records in 0 seconds
```

Report of the Apache web server

Monthly Statistics for October 2026	
Total Hits	1243
Total Files	1131
Total Pages	402
Total Visits	138
Total KBytes	18642
Total Unique Sites	57
Avg / Max Hits per Day	155 / 276
Avg / Max Files per Day	141 / 252
Avg / Max Pages per Day	50 / 93
Avg / Max Visits per Day	17 / 23
Avg / Max KBytes per Day	2330 / 3342

Top URLs by hits

#	Hits	%	URL
1	214	17.2%	/index.html
2	188	15.1%	/css/style.css
3	163	13.1%	/images/logo.png
4	97	7.8%	/login.php
5	76	6.1%	/about.html
6	52	4.2%	/contact.html
7	41	3.3%	/favicon.ico

Top clients

#	Hits	%	Client (site)
1	311	25.0%	192.168.1.25
2	204	16.4%	192.168.1.12
3	142	11.4%	10.0.2.2
4	96	7.7%	192.168.1.30

Hits by response code

HTTP response code	Hits	%
200 - OK	1019	82.0%
304 - Not Modified	112	9.0%
404 - Not Found	74	6.0%
301 - Moved Permanently	21	1.7%
403 - Forbidden	9	0.7%
500 - Internal Server Error	8	0.6%

Comparison of the servers

Server	Records	Total hits	Visits	KBytes	Most requested URL
Apache	1243	1243	138	18642	/index.html
Nginx	856	856	97	12408	/api/status
IIS	522	522	61	7915	/default.aspx
FTP (vsftpd)	64	64	9	41230	/pub/notes.pdf

Inference

- The Apache server is the busiest server and its highest load was on 08-Oct-2026 (276 hits).
- About 82% of the requests were successful (200) and 6% gave 404 errors, which means some links on the site are broken.
- The home page `/index.html` is the most visited page. Image and CSS files form most of the hits, whereas only 402 hits are actual pages.
- The same tool can analyse all the servers because only the log type option (`-F`) changes.

RESULT:

Thus the log files of various types of servers (Apache, Nginx, IIS, FTP) were analysed using the Webalizer tool and the usage statistics such as hits, visits, top URLs, clients and response codes were studied.